

## EE 105 Fall 2025

### Homework 5 – upload to grade scope

(Due November 25, before class, late submission will incur 20% points/day penalty)

**Instructions:** Perform the following tasks based on the circuits and concepts discussed in class. Be sure to show all work where applicable.

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#### 1) Perceptron Model Calculation (Hand Calculation)

- Given a simple perceptron with weights  $w_1 = 2$ ,  $w_2 = -3$ ,  $w_3 = 1.5$  and input values  $x_1 = 1$ ,  $x_2 = 0$ ,  $x_3 = 1$ , calculate the output assuming the threshold is  $0$ . Use the perceptron rule:
  - Output** = 1 if the weighted sum is greater than the threshold, otherwise **0**.

#### 2) Activation Functions (Hand Calculation)

- For a neuron with input  $x = 1.0$  and weight  $w = 2.5$ , calculate the output using:
  - Sigmoid Activation Function:**
  - ReLU Activation Function:**
  - Tanh Activation Function:**

### **3) Design a Neural Network Topology (Conceptual)**

- Design a neural network to classify whether a person should attend a music festival. The inputs are:
  - Weather forecast (good or bad) ◦
  - Availability of tickets (yes or no) ◦
  - Friends attending (yes or no)
- Specify the number of inputs, weights, activation function, and output for this simple model.



#### 4) Smallest Neural Network for MNIST Classification (Python Implementation)

- Your task is to design the smallest possible neural network with a single hidden layer that can classify the MNIST handwritten digits with at least **92% accuracy** for the MNIST TEST dataset.
- Calculate the number of floating point operations (meaning number of multiplications and additions) your neural networks requires to generate a single output. We have gone over this in class, so use the same approach we did there.
- **Requirements:**
  - Use python based on the code developed in the labs to implement your model.
  - The network should contain a **single hidden layer** and output predictions for **10 classes** (digits 0-9).
  - The network should be trained on the **MNIST** dataset.
  - Report the architecture you used, including the size of the hidden layer, what activation functions, and the achieved accuracy.



